

Skip Module Hyperparameter Selection

1 Algorithm

Let F_t denote the grayscale frame at time t . $\mathbf{C}_t \in \{0, 255\}$ is the binary motion mask, where a pixel is declared active if the absolute inter-frame intensity change exceeds threshold τ_d .

Algorithm 1 Frame Difference Detector (FrameDiffDet)

Require: Frames F_t, F_{t-1}

Require: τ_d : diff sensitivity

Ensure: $\mathbf{C}_t \in \{0, 255\}$: binary motion mask

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1: if  $F_{t-1} = \emptyset$  then
2:    $F_{t-1} \leftarrow F_t$ 
3:   return  $\mathbf{0}$ 
4: end if

5:  $\mathbf{C}_t(i, j) \leftarrow \begin{cases} 255 & |F_t(i, j) - F_{t-1}(i, j)| > \tau_d \\ 0 & \text{otherwise} \end{cases}$ 
6:  $F_{t-1} \leftarrow F_t$ 
7: return  $\mathbf{C}_t$ 
```

Let F_t denote the grayscale image at time t , and $\mathbf{M}_t \in [0, M_{\max}]$ the per-pixel *accumulated motion mask*. Δ_t is the *instantaneous delta*, a binary map weighted by α that activates at pixels where the inter-frame intensity change exceeds sensitivity threshold τ_d . $\mathbf{M}_t$ grows by α upon detection and decays by δ each frame, bounded at $M_{\max}$ to prevent indefinite accumulation of stale evidence. A pixel is declared active motion only when $\mathbf{M}_t \geq \tau_m$, suppressing transient noise and single-frame flicker.

Algorithm 2 Accumulation Motion Detector (AccMotionDet)

Require: Frames F_t, F_{t-1} ; accumulated mask $\mathbf{M}_{t-1}$

Require: τ_d : diff sensitivity, α : motion increment, τ_m : activation threshold,
 $M_{\max}$: accumulation cap, δ : decay rate

Ensure: $\mathbf{C}_t \in \{0, 255\}$: binary motion mask; $\mathbf{M}_t$: updated accumulated mask

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1: if  $F_{t-1} = \emptyset$  then
2:    $F_{t-1} \leftarrow F_t, \mathbf{M}_{t-1} \leftarrow \mathbf{0}$ 
3:   return  $\mathbf{0}$ 
4: end if

5:  $\mathbf{D}_t \leftarrow |F_t - F_{t-1}|$  ▷ Absolute inter-frame difference
6:  $\Delta_t(i, j) \leftarrow \begin{cases} \alpha & \mathbf{D}_t(i, j) > \tau_d \\ 0 & \text{otherwise} \end{cases}$  ▷ Instantaneous delta
7:  $\mathbf{A}_t \leftarrow \min(\mathbf{M}_{t-1} + \Delta_t, M_{\max})$  ▷ Accumulate and cap
8:  $\mathbf{M}_t \leftarrow \max(\mathbf{A}_t - \delta, 0)$  ▷ Temporal decay

9:  $\mathbf{C}_t(i, j) \leftarrow \begin{cases} 255 & \mathbf{M}_t(i, j) \geq \tau_m \\ 0 & \text{otherwise} \end{cases}$  ▷ Threshold to binary mask
10:  $F_{t-1} \leftarrow F_t$ 
11: return  $\mathbf{C}_t, \mathbf{M}_t$ 
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